

Demonstration of Fonts and Alignment Filter for Pandoc

Nandakumar Chandrasekhar

2026-07-08

Introduction

This document showcases the features of the `fonts-and-alignment` Pandoc Lua filter. Each example displays the required Markdown syntax alongside its rendered output, ensuring consistent typography across LaTeX, Typst, and HTML formats.¹

The extension leverages Pandoc's `fenced_divs` and `bracketed_spans` extensions. If these are disabled, the filter will issue a terminal warning and render the elements as raw text instead of applying the styles.

Bracketed Spans and Fenced Divs Invocations

This filter follows Pandoc's standard conventions for container-based styling: use Bracketed Spans for inline text and Fenced Divs for block-level content. For further details on these syntax standards, please refer to [Pandoc User's Guide: Divs and Spans](#).

```
[This is a line of text in extra extra large font.]{.pfa-size-2xl}
```

This is a line of text in extra extra large font.

```
::: {.pfa-size-2xl}
```

This multiline block showcases the filters's capability to apply
↪ extra extra large font to an entire section of text by wrapping
↪ the content in a Fenced Div.

```
:::
```

This multiline block showcases the filters's capability to apply extra extra large font to an entire

¹HTML output requires the companion `fonts-and-alignment.css` stylesheet.

section of text by wrapping the content in a Fenced Div.

Font Sizing in Bracketed Spans

The extension provides nine predefined sizing hooks to scale text relative to the document's base/normal font size. The *LaTeX Reference* column is provided for descriptive purposes, as these labels are not native to Typst or HTML/CSS.

LaTeX Reference	Syntax	Output
tiny	<code>[Sample]{.pfa-size-3xs}</code>	Sample
scriptsize	<code>[Sample]{.pfa-size-2xs}</code>	Sample
footnotesize	<code>[Sample]{.pfa-size-xs}</code>	Sample
small	<code>[Sample]{.pfa-size-s}</code>	Sample
normal	<code>[Sample]{.pfa-size-normal}</code>	Sample
large	<code>[Sample]{.pfa-size-l}</code>	Sample
Large	<code>[Sample]{.pfa-size-xl}</code>	Sample
LARGE	<code>[Sample]{.pfa-size-2xl}</code>	Sample
huge	<code>[Sample]{.pfa-size-3xl}</code>	Sample

Font Sizing in Fenced Divs

The same sizing classes can also be applied to Fenced Divs, allowing an entire block of text to be rendered at a specific font size.

Extra Small Font Size

```
::: {.pfa-size-xs}
This paragraph renders at the *extra small* font size.
:::
```

This paragraph renders at the *extra small* font size.

Normal Font Size

```
::: {.pfa-size-normal}
```

This paragraph renders at the **normal** font size.

```
:::
```

This paragraph renders at the *normal* font size.

Large Font Size

```
::: {.pfa-size-l}
```

This paragraph renders at the **large** font size.

```
:::
```

This paragraph renders at the *large* font size.

Font Weights, Shapes, and Families in Bracketed Spans

The extension provides predefined typographic styles for Bracketed Spans. These styles are mapped to equivalent rendering properties across all formats to ensure consistent output, regardless of the underlying engine.

Typographic Style	Syntax	Output
Normal (weight)	<code>[Sample]{.pfa-weight-normal}</code>	Sample
Medium (weight)	<code>[Sample]{.pfa-weight-medium}</code>	Sample
Bold (weight)	<code>[Sample]{.pfa-weight-bold}</code>	Sample
Emphasis (style)	<code>[Sample]{.pfa-style-emph}</code>	<i>Sample</i>
Upright (style)	<code>[Sample]{.pfa-style-upright}</code>	Sample
Italic (style)	<code>[Sample]{.pfa-style-italic}</code>	<i>Sample</i>
Slanted (style)	<code>[Sample]{.pfa-style-slanted}</code>	<i>Sample</i>
Small Caps (style)	<code>[Sample]{.pfa-style-smallcaps}</code>	SAMPLE
Serif (family)	<code>[Sample]{.pfa-family-serif}</code>	Sample
Sans-Serif (family)	<code>[Sample]{.pfa-family-sans}</code>	Sample
Monospace (family)	<code>[Sample]{.pfa-family-mono}</code>	Sample

Font Weights, Shapes, and Families in Fenced Divs

These same typographic classes can also be applied to Fenced Divs, allowing an entire block of text to adopt a particular visual style.

Bold Weight

```
::: {.pfa-weight-bold}
This paragraph is in bold weight.
:::
```

This paragraph is in bold weight.

Small Caps

```
::: {.pfa-style-smallcaps}
This paragraph is in small caps.
:::
```

THIS PARAGRAPH IS IN SMALL CAPS.

Sans-Serif Family

```
::: {.pfa-family-sans}
This paragraph is in sans-serif.
:::
```

This paragraph is in sans-serif.

Text Casing Transformations

Unlike purely visual styling in HTML/CSS—where `text-transform` only changes how text is rendered—these casing transformations operate at the [Pandoc Abstract Syntax Tree \(AST\)](#) level and modify the actual document content. This ensures that uppercase and lowercase conversions are preserved consistently across all output formats and remain intact when copied into other applications.

Casing Transformations in Bracketed Spans

Casing	Syntax	Output
Uppercase	<code>[Sample]{.pfa-case-upper}</code>	SAMPLE
Lowercase	<code>[SAMPLE]{.pfa-case-lower}</code>	sample

Casing Transformations in Fenced Divs

Lowercase

```

::: {.pfa-case-lower}
THIS PARAGRAPH IS TRANSFORMED TO LOWERCASE.
:::

```

this paragraph is transformed to lowercase.

Uppercase

```

::: {.pfa-case-upper}
this paragraph is transformed to uppercase.
:::

```

THIS PARAGRAPH IS TRANSFORMED TO UPPERCASE.

Colors

This section describes how to apply color and spacing using the `pfa-color`, `pfa-bg-color`, and `pfa-padding` attributes within Bracketed Spans and Fenced Divs.

Color Fundamentals

The filter normalizes color inputs to ensure they render identically across LaTeX, Typst, and HTML.

Flexible Color Terminology

For solid colors, the filter is case-insensitive and accepts most naming conventions. All examples below resolve to the CSS3 `mediumvioletred`:

Naming Convention	Syntax	Output
Lowercase	<code>[Color]{pfa-color="mediumvioletred"}</code>	Color
Title Case	<code>[Color]{pfa-color="Medium Violet Red"}</code>	Color
Kebab Case	<code>[Color]{pfa-color="medium-violet-red"}</code>	Color
Snake Case	<code>[Color]{pfa-color="medium_violet_red"}</code>	Color
Camel Case	<code>[Color]{pfa-color="mediumVioletRed"}</code>	Color
Pascal Case	<code>[Color]{pfa-color="MediumVioletRed"}</code>	Color
Screaming Snake Case	<code>[Color]{pfa-color="MEDIUM_VIOLET_RED"}</code>	Color

Input Types

Input Type	Syntax	Output
CSS3 Named	<code>[Sample]{pfa-color="crimson"}</code>	Sample
Hex Full	<code>[Sample]{pfa-color="#2E8B57"}</code>	Sample
Hex Shorthand ²	<code>[Sample]{pfa-color="#666"}</code>	Sample

Compatibility Note: To ensure cross-platform stability, use standard CSS3 named colors or hex codes. Avoid LaTeX-specific numbered variants (e.g., `LightBlue3`), as these are not supported in web browsers or CSS.

Color Mixing

The extension supports LaTeX's `xcolor` percentage-mixing syntax for tints, shades, and blends. This is the standardized syntax for this extension because it maps reliably

²Three-digit shorthand expands by duplicating each single hexadecimal digit per RGB channel (e.g., `#666` expands to `#666666`).

across Typst, LaTeX, and HTML (which leverages the native CSS `color-mix()` function).

Supported Mixing Methods

- **Tinting (Blending with White):**
 - Syntax: `BaseColor!Percentage`
 - Example: `Maroon!40` (Yields 40% Maroon and 60% white)
- **Shading (Blending with Black):**
 - Syntax: `BaseColor!Percentage!black`
 - Example: `MediumVioletRed!80!black` (Yields 80% MediumVioletRed and 20% black)
- **Two-Color Mix:**
 - Syntax: `BaseColor!Percentage!MixColor`
 - Example: `RoyalBlue!50!ForestGreen` (Yields a 50/50 mix of both colors)

Syntax Examples

Mixing Type	Syntax	Output
Tint (40% Base)	<code>[Tinted]{pfa-color="Maroon!40"}</code>	Tinted
Shade (80% Base)	<code>[Shaded]{pfa-color="MediumVioletRed!80!black"}</code>	Shaded
Mix (50/50)	<code>[Mixed]{pfa-font-color="RoyalBlue!50!ForestGreen"}</code>	Mixed

Important Compatibility Rules

- **Strictly Use LaTeX Syntax:** Always use the `BaseColor!Percentage` pattern. This is the only syntax guaranteed to translate correctly across all three output formats.
- **Avoid Typst Native Syntax:** If you use native Typst color syntax (e.g., `color.mix()`), the filter will pass it directly to Typst. It will not be translated for LaTeX or HTML, and those formats will likely receive a fallback color.
- **Binary Mixing Only:** You may blend a maximum of two colors. Multi-color mixing (e.g., `Red!30!Blue!30!Green`) is not supported.
- **Casing Matters:** Because mixing strings are passed directly to LaTeX, casing must be exact:
 - Use lowercase for [core LaTeX colors](#) (e.g., `black`, `red`).

- Use PascalCase for **CSS3 named colors** (e.g., **RoyalBlue**, **MediumVioletRed**).

Styling Elements

Foreground and Background

Use **pfa-color** for text and **pfa-bg-color** for container backgrounds.

```
::: {pfa-color="DarkSlateGrey" pfa-bg-color="lightsteelblue"  
  ↪  pfa-padding="1em"}
```

This block has a default text color and a background fill.

```
[This span overrides the text color.]{pfa-color="tomato"}
```

```
:::
```

This block has a default text color and a background fill. **This span overrides the text color.**

Layout Spacing (**pfa-padding**)

The pfa-padding attribute controls internal spacing using standard CSS shorthand.

- **One value (1em):** Applied to all four sides evenly.
- **Two values (1em 0.5em):** The first value applies to the top and bottom; the second applies to the left and right.
- **Three values (1em 0.5em 2em):** The first value is the top, the second applies to both the left and right, and the third is the bottom.
- **Four values (1em 0.5em 2em 0.2em):** Applied in clockwise order: top, right, bottom, and left.

Supported Units: Only **pt**, **em**, and **ex** are supported to guarantee stability.

Typst Named Palette

You can access Typst's native predefined colors by prefixing the color name with **typst**.

- **Syntax:** `[Sample]{pfa-bg-color="typstmaroon" pfa-color="white" pfa-padding="0.2em"}`
- **Output:** **Sample**

Text Alignment within Fenced Divs

The `pfa-align-*` classes control text alignment within a Fenced Div while explicitly preserving hard line breaks (`\`). These utilities map directly to native LaTeX commands (`\raggedright`, `\centering`, and `\raggedleft`) in PDF output and equivalent CSS behaviors in HTML, making them ideal for formatting poetry, lyrics, or multi-line blocks.

Left-aligned Text

```
::: {.pfa-align-left}
This block of text is *left-aligned*.
:::
```

This block of text is *left-aligned*.

Center-aligned Text

```
::: {.pfa-align-center}
This block of text is *center-aligned*.
:::
```

This block of text is *center-aligned*.

Right-aligned Text

```
::: {.pfa-align-right}
This block of text is *right-aligned*.
:::
```

This block of text is *right-aligned*.

Line Break Preservation

```
::: {.pfa-align-center}
This block of text\
is *center-aligned*\
while preserving explicit line breaks.
:::
```

This block of text
is *center-aligned*
while preserving explicit line breaks.

Combining Multiple Classes

Multiple typographic utilities can be combined within the same element. Font, size, color, alignment, and decoration classes are designed to compose independently and can be applied together to both Bracketed Spans and Fenced Divs.

Class names are space-separated within `{ }` following Pandoc attribute syntax. In addition to classes, key-value attributes such as `pfa-color` may be included alongside class definitions.

Bracketed Span Composition

Style	Syntax	Output
Bold Sans-Serif	<code>[Sample]{.pfa-wight-bold .pfa-family-sans .pfa-size-l pfa-color="red"}</code>	Sample
Italic Monospace	<code>[Sample]{.pfa-style-italic .pfa-family-mono .pfa-size-s}</code>	<i>Sample</i>
Small Caps, Colored	<code>[Sample]{.pfa-style-smallcaps pfa-color="forestgreen"}</code>	SAMPLE

Fenced Div Composition

Multiple utilities can also be applied to Fenced Divs to control alignment, typography, and color simultaneously. The resulting block inherits all specified styles while preserving Pandoc's standard attribute behavior.

```
::: {.pfa-align-center .pfa-family-sans .pfa-weight-bold .pfa-size-l
↪  pfa-color="midnightblue"}
A centered, bold, sans-serif, large, midnight-blue Fenced Div
↪  demonstrating multiple combined utilities from the filter.
:::
```

**A centered, bold, sans-serif, large, midnight-blue Fenced Div
demonstrating multiple combined utilities from the filter.**

Legacy Aliases (Deprecated)

Warning: Legacy aliases are deprecated. They are retained strictly for backward compatibility and will be entirely removed in the next major release. New documents

should use the `pfa-*` namespaces going forward.

Font Sizing Aliases

Legacy Class	Modern Class
<code>xsmall</code>	<code>pfa-text-xs</code>
<code>small</code>	<code>pfa-text-s</code>
<code>normal</code>	<code>pfa-text-normal</code>
<code>large</code>	<code>pfa-text-l</code>
<code>xlarge</code>	<code>pfa-text-xl</code>
<code>xxlarge</code>	<code>pfa-text-2xl</code>
<code>huge</code>	<code>pfa-text-3xl</code>

Font Weight, Shape, and Family Aliases

Legacy Class	Shorthand	Modern Class
<code>bold</code>	<code>bf</code>	<code>pfa-font-bold</code>
<code>emphasis</code>	<code>em</code>	<code>pfa-font-emphasis</code>
<code>italic</code>	<code>it</code>	<code>pfa-font-italic</code>
<code>medium</code>	<code>md</code>	<code>pfa-font-medium</code>
<code>monospace</code>	<code>tt</code>	<code>pfa-font-mono</code>
<code>normalfont</code>	<code>nf</code>	<code>pfa-font-normal</code>
<code>sans</code>	<code>sf</code>	<code>pfa-font-sans</code>
<code>serif</code>	<code>rm</code>	<code>pfa-font-serif</code>
<code>slanted</code>	<code>sl</code>	<code>pfa-font-slanted</code>
<code>smallcaps</code>	<code>sc</code>	<code>pfa-font-smallcaps</code>
<code>upright</code>	<code>up</code>	<code>pfa-font-upright</code>

Alignment Aliases

Legacy Class	Modern Class
<code>centering</code>	<code>pfa-align-center</code>
<code>raggedleft</code>	<code>pfa-align-right</code>
<code>raggedright</code>	<code>pfa-align-left</code>

Text Decoration Aliases

Legacy Class	Shorthand	Modern Class
<code>uline</code>	<code>u</code>	<code>pfa-text-underline</code>
<code>uuline</code>	<code>uu</code>	<code>pfa-text-underline-double</code>
<code>dashuline</code>	<code>dau</code>	<code>pfa-text-underline-dashed</code>
<code>dotuline</code>	<code>dou</code>	<code>pfa-text-underline-dotted</code>
<code>uwave</code>	<code>uw</code>	<code>pfa-text-underline-wave</code>
<code>sout</code>	<code>so</code>	<code>pfa-text-strikeout</code>

Removed Classes

The following legacy classes were removed in version 2.0.0 and are no longer recognized by the filter. Existing documents should be updated to use the replacement classes shown below.

Removed Class	Shorthand	Removed In	Replacement
<code>center</code>	—	2.0.0	<code>pfa-align-center</code>
<code>flushleft</code>	—	2.0.0	<code>pfa-align-left</code>
<code>flushright</code>	—	2.0.0	<code>pfa-align-right</code>
<code>xout</code>	<code>xo</code>	2.0.0	<code>pfa-text-strikeout</code>